

Connect through a Proxy

The Networking Properties

You have to set the following properties:

```
http.proxyHost (default: <none>)
http.proxyPort (default: 80 if http.proxyHost specified)
http.nonProxyHosts (default: <none>)
```

NOTE: *proxyHost*, *proxyPort* are deprecated. You have to prefix them with "http.".

NOTE: Those properties are documented here:

<http://java.sun.com/javase/6/docs/technotes/guides/net/properties.html>.

You can set the required properties when starting the JVM for a JAVA application from the command line:

```
java -Dhttp.proxyHost=myproxyserver.com -
Dhttp.proxyPort=80 MyJavaApp
```

Or in your source :

```
System.setProperty("http.proxyHost", "myProxyServer.com");
System.setProperty("http.proxyPort", "80");
```

Since **Java 1.5** you can also pass a `java.net.Proxy` instance to the `openConnection()` method:

```
//Proxy instance, proxy ip = 123.0.0.1 with port 8080
Proxy proxy = new Proxy(Proxy.Type.HTTP, new
InetSocketAddress("123.0.0.1", 8080));
URL url = new URL("http://www.yahoo.com");
URLConnection uc =
(HttpURLConnection)url.openConnection(proxy);
uc.connect();
```

String page;

```
StringBuffer tmp = new StringBuffer();
BufferedReader in = new BufferedReader(new
InputStreamReader(uc.getInputStream()));
while ((line = in.readLine()) != null){
    page.append(line + "\n");
}
System.out.println(page);
```

so you don't need to set system properties.

You can use the default PROXY as defined by your networking settings.

```
System.setProperty("java.net.useSystemProxies", "true");

List l = null;
try {
    l = ProxySelector.getDefault().select(new
    URI("http://www.yahoo.com"));
}
catch (URISyntaxException e) {
    e.printStackTrace();
}

if (l != null) {
    for (Iterator iter = l.iterator(); iter.hasNext() {
        java.net.Proxy proxy = (java.net.Proxy) iter.next();
        System.out.println("proxy hostname : " +
        proxy.type());
        InetSocketAddress addr = (InetSocketAddress)
        proxy.address();
        if (addr == null) {
            System.out.println("No Proxy");
        }
        else {
            System.out.println("proxy hostname : " +
            addr.getHostName());
            System.out.println("proxy port : " +
            addr.getPort());
        }
    }
}
```

To bypass the PROXY,

```
URL url = new URL("http://internal.server.local/");
URLConnection conn = url.openConnection(Proxy.NO_PROXY);
```

Proxy and Username/Password

You might need to identify yourself to the proxy server.

One way is to use the HTTP property "Proxy-Authorization" with a username:password base64 encoded.

```
System.setProperty("http.proxyHost", "myProxyServer.com");
System.setProperty("http.proxyPort", "80");
URL url=new URL("http://someserver/somepage");
URLConnection uc = url.openConnection ();
String encoded = new String
    (Base64.base64Encode(new
    String("username:password").getBytes()));
uc.setRequestProperty("Proxy-Authorization", "Basic " +
```

```
encoded);  
uc.connect();
```

NOTE: For a base64 function, see this [How-to](#).

The following example dumps the content of a URL but before we identify ourselves to the proxy.

```
import java.net.*;  
import java.io.*;  
  
public class URLUtils {  
    public static void main(String s[]) {  
        URLUtils.dump("http://www.yahoo.com");  
        System.out.println("*****");  
        URLUtils.dump("https://www.paypal.com");  
        System.out.println("*****");  
    }  
  
    public static void dump(String URLName){  
        try {  
            DataInputStream di = null;  
            FileOutputStream fo = null;  
            byte [] b = new byte[1];  
  
            // PROXY  
  
            System.setProperty("http.proxyHost", "proxy.mydomain.local")  
            ;  
            System.setProperty("http.proxyPort", "80") ;  
  
            URL u = new URL(URLName);  
            HttpURLConnection con = (HttpURLConnection)  
u.openConnection();  
            //  
            // it's not the greatest idea to use a sun.misc.*  
class  
            // Sun strongly advises not to use them since they  
can  
            // change or go away in a future release so beware.  
            //  
            sun.misc.BASE64Encoder encoder = new  
sun.misc.BASE64Encoder();  
            String encodedUserPwd =  
  
encoder.encode("mydomain\\MYUSER:MYPASSWORD".getBytes());  
            con.setRequestProperty  
                ("Proxy-Authorization", "Basic " +  
encodedUserPwd);  
            // PROXY -----  
  
            di = new DataInputStream(con.getInputStream());
```

```

        while(-1 != di.read(b,0,1)) {
            System.out.print(new String(b));
        }
    }
    catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

With JDK1.2, the [java.net.Authenticator](#) can be used to send the credentials when needed.

```

public static void dump(String URLName){
    try {
        DataInputStream di = null;
        FileOutputStream fo = null;
        byte [] b = new byte[1];

        // PROXY

        System.setProperty("http.proxyHost", "proxy.mydomain.local");
        System.setProperty("http.proxyPort", "80") ;

        Authenticator.setDefault(new Authenticator() {
            protected PasswordAuthentication
            getPasswordAuthentication() {
                return new

                PasswordAuthentication("mydomain\\username", "password".toCh
                arArray());
            }
        });

        URL u = new URL(URLName);
        HttpURLConnection con = (HttpURLConnection)
        u.openConnection();
        di = new DataInputStream(con.getInputStream());
        while(-1 != di.read(b,0,1)) {
            System.out.print(new String(b));
        }
    }
    catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

Bypass a Proxy

In intranet environment, you may need to bypass the proxy server and go directly to the http server.

The *http.nonProxyHosts* property indicates the hosts which should be connected too directly and not through the proxy server. The value can be a list of hosts, each seperated by a |, and in addition a wildcard character (*) can be used for matching.

```
java.exe
```

```
-
```

```
Dhttp.nonProxyHosts="*.mycompany.com|*.mycompany.local|localhost"
```

```
MyClass
```